



Immunomodulatory effects of Abnormal Savda Munsiq, a traditional Uighur medicine, on the combined stress mice

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ABSTRACT

Objective: To investigate the immunomodulatory effects of ASMq, a herbal preparation used in Traditional Uighur Medicine, on the combined stress mice.

Materials and methods: The combined stress was induced on mice by application of the electric-foot shock in a cold–dry environment and three different doses of ASMq were orally administered to the animals for 14 days. The effect of ASMq on the immune apparatus weight index, lymphocytes proliferation, serum levels of the cytokines, immunoglobulins, hemolysin and NK cells as well as the phagocytic activity of the macrophages were evaluated.

Results: Oral administration of ASMq was found to increase the thymus and spleen indices, lymphocytes proliferation induced by Con A and LPS, the percentage of CD4⁺ in thymus, spleen and peripheral blood and restore the CD4⁺/CD8⁺ ratio. The serum concentrations of INF- β , IL-2, IL-6, IgG, NK cells and hemolysin were also increased. The macrophage phagocytic activity was also enhanced.

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1. Introduction

Munziq is one of the herbal preparations widely used in Traditional Uighur Medicine (TUM) for the prevention and treatment of different syndromes. According to Traditional Uighur Medicine *hilit* theory, the four essential body fluids Kan, Belghem, Sapra and Savda should be in balance to keep the body healthy. Alteration of one or more of these fluids can affect the human metabolism and consequently cause diseases (Upur et al., 2004). Thus, the syndromes generally described in TUM can be divided into four main categories: abnormal Savda, abnormal Sapra, abnormal Belghem and abnormal Kan syndromes. According to TUM, Abnormal Savda Syndrome is described as a condition which is neither completely healthy nor ill and if it is not prevented in time, it will develop complex chronic diseases. Thus, in the ancient theory of TUM, the treatment of this condition was recognized as more important than treating the end point disease (Upur and Yusup, 2003). In addition, TUM theory also describes the cold and dry environment, psychological and/or emotional stressors as causative factors of Abnormal Savda Syndrome. In spite of the fact that the physiology of Abnormal Savda Syndrome is not yet fully understood, it is assumed that

the syndrome is associated with the immune deregulation. Thus, in view of modern medicine, Abnormal Savda Syndrome can be equivalent to the situation described for the complex chronic diseases such as diabetes, cancer, cardiovascular diseases and chronic asthma.

A human body always receives different stressors simultaneously and the negative effects of these stressors are associated with the increasing risks of developing diseases such as diabetes, cancer and cardiovascular diseases (Jonas and Lando, 2000; Sephton and Spiegel, 2003; Eggers, 2007). Though, the network of the three main systems (nervous, endocrine and immune systems) are of the clinical main interests, there are increasing evidences that the immune system is involved in the development of several chronic illnesses (Bartolomucci, 2005). Also, it is well recognized that the persistent activation of the hypothalamic–pituitary–adrenal (HPA) axis in chronic stress response and in depression can also impair the immune response and thus, contributes to the development and progression of certain types of chronic diseases (Irwin, 2008). On the hand, there are also evidences indicating that various cellular and molecular immunological factors are compromised under the chronic stress and these factors are found to play an important role in the onset and progression of chronic diseases (Leonard, 2000; Maddock and Pariente, 2001). Thus, a better insight on the bidirectional communication between the neuro-endocrine and the immune systems can contribute to a new strategy for their clinical treatments. As the immune system is involved in

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different processes of chronic diseases, it is the target of our study to modulate its function in order to decrease their risks.

Abnormal Savda Munsig (ASMq) has been previously investigated for its capacity to decrease biological markers of oxidative stress in man (Upur et al., 2004) as well as to protect mitochondrial membranes against OH-induced oxidative damage (Yusup et al., 2004). The aqueous extract of ASMq was also found to exert its inhibition of the proliferation and viability of human hepatoma (HepG2) cells (Yusup et al., 2005). However, there is no study describing the effects of ASMq on the immune system so far. Consequently, we have investigated the effects of ASMq, according to its ethnopharmacological use (treatment and prevention of chronic diseases), on the stress-induced immune dysfunction mice produced by cold/dry environment and electrical foot-shock (psychological or emotional). We have found in this study that ASMq was able to clearly restore the immune function of the combined stress mice. The results of this study are very important to warrant the claim of the traditional use of ASMq prescription.

2. Materials and methods

2.1. Reagents

Fluorescein isothiocyanate (FITC) anti-mouse CD4 and phycoerythrin (PE) anti-mouse CD8 were purchased from Biolegend (USA) and phycoerythrin (PE) anti-mouse NK1.1 was from Bioscience (USA). Bovine serum albumin (BSA), Hanks' balanced salt solution (HBSS), concanavalin A (Con A) and lipopolysaccharide (LPS) were purchased from Sigma Aldrich (USA). RPMI 160 was purchased from Gibco BRL (NY, USA). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT), bovine serum albumin were from Sigma (St. Louis, USA). Sheep red blood cell (SRBC) serum was provided by Xinjiang Autonomous Region Medical Experiment Animal Center; mouse IL-2, IL-6, INF- β , IgM, IgG, IgA ELISA kit were purchased from R&D System (Minneapolis, USA).

2.2. Preparation of the aqueous extract of ASMq

The best quality commercial ingredients of the ASMq prescription (*Adiantum capillus-veneris* L. whole plant, *Alhagi pseudodolhagi* (Bieb.) Desv. branch secretion, *Anchusa ilalica* Retz. whole plant, *Cordia dichotoma* G. Forst. fruit, *Euphorbia maculata* L. whole plant, *Foeniculum vulgare* Mill. fruit, *Glycyrrhiza glabra* L. rhizome, *Lavandula angustifolia* Mill. aerial part, *Melissa officinalis* L. whole plant, *Ziziphus jujuba* Mill. Fruit) were purchased from Xinjiang Autonomous Region Traditional Uighur Medicine Hospital in Urumqi and were authenticated by the Associate Chief Pharmacist of the Uighur Medicine Preparation Center of the hospital. The ASMq was prepared according to the recipe recommended by the Chinese Pharmacopoeia concerning traditional Uighur preparation (Upur et al., 2004). Briefly, all the herbs were chopped into small pieces and ground. The powdered material (1 kg) was macerated in warm distilled water (10 L) for 12 h. The solution was then boiled for 30 min and left to stand for 1 h before filtration. The filtrate was then concentrated to semisolid mass under reduced pressure and dried at 60 °C under vacuum condition. The yield of the extract was 29.4% (w/w) of the dried plant materials. The extract was dissolved in 0.5% Sodium carboxymethyl cellulose (CMCNa) solution and administered to the mice according to the experimental dosages.

2.3. Animal

The pathogen free ICR male mice weighing 18–20 g (Xinjiang Medical University Animal Center, Urumqi, China) were under con-

dition of controlled temperature (24 °C) and illumination (12 h light cycle) and were maintained on laboratory standard diet and water freely. The animals were divided randomly into five groups of 12 mice. The mice of normal group (Gr. I) were housed and fed in the normal condition.

The combined stress was induced on mice by application of the electric-foot shock in a cold-dry environment. The combined stress model group (Gr. II) received the repeated electric foot-shock (volume 20–30 V; interval 0.2–0.5 s during 20 min/day) in the electric foot-shock instrument before being housed in the climatic cabinet from 11:00 am to 9:00 pm (at 6 °C and 25–32.8% of relative humidity). The experimental procedures were carried out according to the guidelines for the animal care and use approved by the committee of Traditional Uighur Medicine Institute, Xinjiang Uighur Autonomous region.

2.4. Drug administration

0.5% CMCNa solution (20 ml/kg, b.w.) was administered intragastrically to the normal (Gr. I) and the combined stress model (Gr. II) mice once a day during 14 days. Another group of mice received ASMq solution orally at the dose of 2.53 g/kg (Gr. IIIa), 5.06 g/kg (Gr. IIIb) and 10.12 g/kg (IIIc.), respectively between 11:00 am and 12:00 daily during 14 days. The doses of ASMq were calculated according to the standard dose of *Curcuma longa* extract used in traditional Chinese medicine clinical practice. Food was withdrawn from the animals 2 h prior to drug administration but water was allowed freely.

2.5. Effects of ASMq on the immune apparatus index

Mice were euthanized by CO₂ inhalation. Body weights of mice were measured regularly; the weights of spleen and thymus were measured after being excised from the animals. The thymus and spleen indices are expressed as milligrams of the thymus or spleen per 10 g of body weight, respectively.

2.6. Lymphocyte proliferation assay

Lymphocyte proliferation was determined by MTT assay. Mice were sacrificed by cervical dislocation and spleen was removed under sterile condition. Splenocytes were obtained by gentle disruption of the spleen of mice and filtration via a 40- μ Nylon cell strainer (Falcon, NJ, USA). The erythrocytes were lysed with 0.38% NH₄Cl-Tris buffer (pH 7.4), while the remaining cells were resuspended in RPMI-1640 with 10 mM Hepes, 10% foetal bovine serum (FBS), 100 mg/ml streptomycin and 100 IU/ml penicillin. After two washes in HBSS, the cells were resuspended in RPMI-1640 with 10% heated inactivated FBS. The viability of splenocytes tested by trypan blue dye exclusion was always more than 95%. A total of 2×10^6 cell/L lymphocytes cells in 200 μ L of RPMI-1640, supplemented with 10% FBS were stimulated with 5 mg/ml Con A or 50 mg/ml LPS respectively, and co-cultured with test samples in 96-well plates for 48 h at 37 °C in a humidified atmosphere of 5% CO₂; 10 μ L of MTT (1 mg/ml) was added to the cells and oscillated for 1 min on oscillator, followed by incubation for 1 h. After centrifugation, the medium was removed and 200 μ L of DMSO were added to each well. The absorbance (A) was measured on Micro well Reader at 490 nm. The experiments were performed in triplicates and the results were described as average of A.

2.7. Thymocyte and splenocyte subsets detection

Lymphocyte subsets of spleen and thymus were analyzed in duplicate to determine whether there were differences in the ratio

of T cells. Indirect immunofluorescence was performed using fluorescein isothiocyanate (FITC) anti-mouse CD4 and phycoerythrin (PE) anti-mouse CD8. Spleen and thymus were removed under sterile condition and lymphocytes were isolated and collected by routine method. The cells were suspended in RPMI-1640 medium at a concentration of 1×10^6 cells/L. The cultures were incubated at 37 °C, 5% CO₂ for 30–45 min, after that the cells were labelled with FITC or PE for 20 min. The cell suspension was passed through a #20 metal sieve and washed with phosphate-buffered saline (PBS) containing 1.5% FBS. Immunofluorescence was determined with flow cytometry using a Becton Dickinson FAC star Plus analyzer. A fluorescence histogram of at least 5000 counts was collected in each sample.

2.8. Peripheral blood lymphocyte subsets detection

From each mouse, 200 µL of blood was collected into an EDTA containing tube. Samples of 100 µL were then stained with FITC or PE for 20 and 15 min respectively, at room temperature in the dark. Red blood cells were lysed with NH₄Cl–Tris and immunofluorescence was determined with flow cytometry using a Becton Dickinson FAC star Plus analyzer.

2.9. Peritoneal macrophage neutral red dye uptake assay

Cell culture: Peritoneal cells were harvested 3 days after intraperitoneal injection of 2 ml of starch solution (1% concentration) and isolated using 10 ml of HBSS containing heparin 10 U/ml. The cells were then distributed in RPMI-1640 medium supplemented with 10% FCS, penicillin (100 U/ml) and streptomycin (100 µg/ml), incubated for 3 h (at 37 °C in an atmosphere of 5% CO₂) and washed with HBSS to remove non-adherent cells.

Neutral red uptake assay: Macrophages were incubated at a concentration of 2×10^6 cell/L for 2 h with serum free medium containing neutral red (5 mg/L). The cells were washed quickly with PBS and 0.1 ml of acetic acid solution (1%, v/v) in ethanol (50%, v/v) was added to each well to extract the dye phagocytised by macrophages. After rapid agitation on a microtitre plate shaker, the absorbance (A) was read at 540 nm. Tests were performed in triplicates from three experiments.

2.10. Hemolysin antibody test

On day ninth of drug administration, the mice were immunized by intraperitoneal injection of 0.2 ml of 10% SRBC. Five days after immunization, serum was collected to measure the concentration of hemolysin antibody.

2.11. Measurement of IL-2, IL-6, INF-β, IgM, IgG and IgA concentrations

On the day fourteenth of drug administration, the blood was collected and centrifuged at 4 °C; the serum was stored at –80 °C before assay. The IL-2, IL-6, INF-β, IgM, IgG and IgA contents were determined by ELISA according to the manufacturer's specification.

2.12. Natural killer cells detection in peripheral blood

Blood was collected from the animal by venipuncture into the EDTA containing tube. 100 µL blood samples were stained by PE anti-mouse NK1.1 for 20 min at room temperature in the dark. Red blood cells were lysed with NH₄Cl–Tris and immunofluorescence was determined with flow cytometry using a Becton Dickinson FAC star Plus analyzer.

Table 1
Effect of ASMq on the weight index of the immune organs.

Mice	Thymus weight index	Spleen weight index
Gr. I	16.21 ± 3.93	56.21 ± 3.93
Gr. II	9.54 ± 5.33*	30.74 ± 5.54*
Gr. IIIa	17.91 ± 6.63***	30.89 ± 6.54
Gr. IIIb	18.51 ± 7.95**	40.28 ± 8.12**
Gr. IIIc	17.42 ± 5.94***	43.36 ± 13.39**

* Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.01$.

** Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.

*** Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.01$.

2.13. Statistical analysis

All values were presented as mean ± S.D. Statistical analysis of the data for multiple comparisons was performed by one-way analysis of variance followed by Duncan's test. For a single comparison, the significance of differences between means was determined by Student's *t*-test. A level of $P < 0.05$ was taken as statistically significant.

3. Results

Table 1 shows that the thymus and spleen indices of the mice in the combined stress model (Gr. II) were decreased when compared to those of the normal group (Gr. I). All three doses of ASMq were found to decrease the degeneration of the thymus, whereas only large and medium doses (5.06 g/kg and 10.12 g/kg b.w.) could protect the spleen as indicated by an increase of the indices of the two immune apparatus when compared to the model group (Gr. II). Oral administration of ASMq was also found to increase lymphocyte proliferation stimulated by Con A and LPS when compared to the model group (**Table 2**). The percentage of CD4⁺ in thymus, spleen and peripheral blood was dramatically decreased in mice under the combined stress condition (Gr. II) when compared to the animals maintained in a normal condition (Gr. I) while that of the CD8⁺ subset was slightly decreased. It was found that not only the percentage of CD4⁺ was significantly increased, but also the CD4⁺/CD8⁺ ratios were restored when the stressed mice were treated with ASMq during 14 days (**Tables 3 and 4**). Oral administration of ASMq was able to restore the phagocytic activity of the macrophages as indicated by the results obtained from neutral red dye uptake assay as well as to increase the serum hemolysin concentration (**Table 5**) in the stress-induced mice. When compared to the normal group (Gr. I). The serum levels of INF-β and IL-2 were decreased significantly while that of IL-6 was increased in the model group (Gr. II). Oral administration of ASMq for 14 days

Table 2
Effect of ASMq on lymphocyte proliferation induced by Con A and LPS.

Mice	Absorbance at 490 nm	
	Con A	LPS
Gr. I	0.723 ± 0.072	0.519 ± 0.095
Gr. II	0.456 ± 0.085*	0.378 ± 0.085*
Gr. IIIa	0.597 ± 0.072**	0.389 ± 0.071
Gr. IIIb	0.628 ± 0.085***	0.419 ± 0.061**
Gr. IIIc	0.709 ± 0.064***	0.466 ± 0.068**

* Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.01$.

** Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.

*** Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.01$.

Table 3
Effect of ASMq on thymocyte and splenocyte subsets.

	Thymocyte subsets (%)			Splenocyte subsets (%)		
	CD4 ⁺	CD8 ⁺	CD4 ⁺ /CD8 ⁺	CD4 ⁺	CD8 ⁺	CD4 ⁺ /CD8 ⁺
Gr. I	27.42 ± 9.44	9.17 ± 2.12	3.13 ± 1.26	26.67 ± 5.48	12.16 ± 2.43	2.43 ± 0.87
Gr. II	18.25 ± 3.74 [*]	16.92 ± 3.53 [*]	1.14 ± 0.39 [*]	15.75 ± 4.67 [*]	10.91 ± 4.01 [*]	1.63 ± 0.79 [*]
Gr. IIIa	21.67 ± 5.39	13.83 ± 3.09	1.42 ± 0.52	16.08 ± 5.82	10.58 ± 2.39	1.65 ± 0.85
Gr. IIIb	26.25 ± 6.12	11.83 ± 2.25 ^{**}	1.89 ± 1.05	21.67 ± 5.39	11.25 ± 4.04	2.15 ± 0.91
Gr. IIIc	24.83 ± 5.84 ^{**}	11.25 ± 4.04 ^{**}	2.15 ± 0.58	24.67 ± 9.33	10.66 ± 3.36 ^{**}	2.67 ± 1.61 ^{**}

^{*} Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.01$.^{**} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.**Table 4**
Effect of ASMq on T-lymphocyte subsets and NK cells in peripheral blood.

	CD4 ⁺	T-lymphocyte subsets		NK
		CD8 ⁺	CD4 ⁺ /CD8 ⁺	
Gr. I	33.72 ± 7.29	8.35 ± 2.07	4.13 ± 1.05	15.06 ± 4.35
Gr. II	22.22 ± 2.57 [*]	8.84 ± 3.03 [*]	2.81 ± 1.03 [*]	8.21 ± 3.58 [*]
Gr. IIIa	33.09 ± 8.11 ^{***}	8.13 ± 3.66	4.84 ± 1.19 ^{***}	9.45 ± 2.34 ^{**}
Gr. IIIb	30.30 ± 4.36 ^{***}	7.48 ± 1.04	4.11 ± 0.76 ^{***}	12.01 ± 3.12 ^{***}
Gr. IIIc	32.38 ± 5.32 ^{***}	8.96 ± 3.48	4.47 ± 1.07 ^{***}	13.78 ± 4.21 ^{**}

^{*} Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.01$.^{**} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.^{***} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.01$.**Table 5**
Effect of ASMq on peritoneal macrophage phagocytic activity and hemolysis production induced by SRBC.

	Macrophage phagocytic activity (A)	Serum hemolysis concentration (HC ₅₀)
Gr. I	0.397 ± 0.082	57.63 ± 7.67
Gr. II	0.113 ± 0.025 [*]	35.63 ± 6.77 [*]
Gr. IIIa	0.139 ± 0.021 ^{**}	41.54 ± 4.77 ^{**}
Gr. IIIb	0.197 ± 0.042 ^{***}	47.35 ± 4.67 ^{***}
Gr. IIIc	0.310 ± 0.035 ^{***}	45.94 ± 9.09 ^{***}

^{*} Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.01$.^{**} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.^{***} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.01$.

was found to increase the serum concentrations of both INF- β and IL-2 but decrease the level of IL-6 of the stressed mice (Gr. IIIa, IIIb and IIIc) (Table 6). Concerning its effect on immunoglobulins, it is interesting to observe that only IgG was increased by the stress condition and all doses of ASMq were found to reduce its level (Table 7).

Table 6
Effect of ASMq on serum concentration of INF- β , IL-2 and IL-6.

	INF- β (ng/ml)	IL-2 (ng/ml)	IL-6 (ng/ml)
Gr. I	26.40 ± 9.88	11.02 ± 3.53	9.80 ± 3.65
Gr. II	7.72 ± 05.28 [*]	4.13 ± 2.33 [*]	14.66 ± 3.62 [*]
Gr. IIIa	11.05 ± 8.11 ^{***}	11.05 ± 3.74 ^{***}	10.31 ± 3.39 ^{**}
Gr. IIIb	18.17 ± 3.45 ^{***}	45 ± 4.12 ^{***}	9.83 ± 3.27 ^{**}
Gr. IIIc	18.18 ± 5.90 ^{***}	9.48 ± 3.26 ^{***}	10.32 ± 2.60 ^{**}

^{*} Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.05$.^{**} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.^{***} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.01$.**Table 7**
Effect of ASMq on serum concentration of IgM, IgG and IgA.

	IgM (ng/ml)	IgG (ng/ml)	IgA (ng/ml)
Gr. I	3.59 ± 1.42	3.51 ± 0.86	10.72 ± 3.05
Gr. II	3.91 ± 1.51	6.74 ± 0.75 [*]	12.05 ± 2.27
Gr. IIIa	4.40 ± 0.93	4.16 ± 1.02 ^{**}	11.17 ± 2.35
Gr. IIIb	4.27 ± 1.09	4.20 ± 0.91 ^{**}	12.28 ± 1.66
Gr. IIIc	3.93 ± 0.68	3.67 ± 1.53 ^{***}	13.27 ± 2.17

^{*} Results are given as means ± S.D., when compared to the model group (Gr. II) $p < 0.01$.^{**} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.05$.^{***} Results are given as means ± S.D., when compared to the normal group (Gr. I) $p < 0.01$.

4. Discussion

The immune system is increasingly found to be involved in the development of several chronic illnesses for which allopathic medicine has provided limited tools for treatment and prevention. On the other hand, the relationship between stress exposures and diseases susceptibility is widely recognized nowadays. Interestingly, the clinical interest focuses on the four major systems (and the related diseases), which appear to be vulnerable to excessive stress-mediator levels: the cardiovascular system (heart failure and hypertension); the immune system (from autoimmunity to cancer and allergies); the metabolic system (obesity); and the nervous system (depression, anxiety). Among these, the immune system is found to be particularly sensitive to stress and chronic stress typically provokes an immune deregulation (Katon et al., 2004; Sephton and Spiegel, 2003). There is good evidence, from both animal and human studies, that the magnitude of stress-associated immune deregulation is large enough to have health implications. It is now widely accepted that stressful events can cause an impact on the health of an individual. It is also well known that acute stress in any form will cause, at the very least, a temporary decrease in functioning of the immune system, while chronic stress will result in continued decline in the function of the immune system (Padgett and Glaser, 2003; Schaaf and Cidlowski, 2002). It is also suggested that psychological stress is associated with increased immune deregulation and oxidative damage, and thus, long-term exposure to psychological stressors may enhance the risk of many diseases (Liu and Wang, 2005). The immune deregulation may also lead to the development of complex chronic diseases such as cancer (Bultz and Carlson, 2005), cardiovascular diseases (Busse et al., 2001) and diabetes (Sepa et al., 2005). In this perspective, psychological chronic combined stress-induced (cold-dry environment and electronic foot-shock) pathophysiology is similar to the so-called Abnormal Savda Syndrome (sub-healthy or pre-diseases), condition defined in the ancient theory of TUM.

In this study, we have developed a simple model for studying the effect of ASMq, using cold and/or dry environment conditions (chronic stress) and electric foot-shock (psychological and emotional) to induce the stress in mice in according with TUM theory

which regards the cold and dry environment, psychological and/or emotional stressors as causative factors of Abnormal Savda Syndrome.

ASMq is a traditional Uighur recipe comprising ten herbal drugs and is used in Southern Xinjiang Province of China not only by Uighur medical doctors for treating Abnormal Savda Syndrome but also by ordinary people for self-medication. Chemically, these herbal medicines are all rich in phenolics, flavonoids, saponin glycosides as well as polysaccharides (Shimizu et al., 1991; Li et al., 2000; Mojab et al., 2003; Amakura et al., 1997; Alimova et al., 2007). Pharmacologically, it is widely established that the dual-direction immunomodulatory effects of plant flavonoids are of profound significance for the immune deregulation diseases (Gábor, 1986; Mookerjee, 1986). On the other hand, saponin is also an active immunomodulator, stimulating the specific and non-specific immunities (Rajput et al., 2007). Ultimately, polysaccharides are widely known to possess the effect of modulating immuno-function by multi-ways, multi-targets and many targets such as promoting proliferation and differentiation of lymphocytes, secreting various lymphokines, modulating the balance of neuroendocrine immunomodulation network (Ooi and Liu, 2000).

The results of our study showed that oral administration of ASMq preparation (2.53 g/kg, 5.06 g/kg and 10.12 g/kg b.w.) to the combined stress mice has resulted in an observed regulation effect on both cell and humoral immunities. ASMq was found to increase the immune apparatus weight indices, indicating that the preparation can restore the immunity function, i.e. it can delay the degeneration of thymus and spleen to some extent in the combined stress mice. Interestingly, Zhao et al. (2006) have reported isolation of a pectic polysaccharide of polygalacturonan type from the fruits of *Ziziphus jujuba*, which induced a rat spleen cells proliferation in a dose dependent manner. In the same time, ASMq was found to significantly enhance the proliferation of splenocytes better than Con A and LPS, the well-known mitogens which stimulate different lymphocytes, B and T-cells. Equally important is the helper T lymphocytes (Th) homeostasis which is crucial in orchestrating the appropriate cytokine responses and hence remains one of the targets for immunomodulation. These subsets involve fundamentally different and opposing effectors functions, and extreme of either of these can lead to diseases. Thus, the optimal immunotherapy should restore or maintain a well balanced Th1 and Th2 response, suited to the immune challenge (Mosmann and Coffman, 1989). The study showed that the percentages of CD4⁺ in thymus, spleen and peripheral blood were increased, and the CD4⁺/CD8⁺ ratios were resumed with administration of ASMq. Moreover, treatment of ASMq has resulted in a rapid recovery of CD4⁺ counts under immunosuppressed conditions, indicative of immune restorative activity of this preparation.

On the other hand, complex cellular and inflammatory interactions are involved in the progress of vascular chronic diseases. Endothelial cells, upon exposure to cytokines, can undergo profound alterations of the function that involve gene expression and *de novo* protein synthesis. The functional reprogramming of endothelial cells by cytokines is of importance, especially in patients with chronic inflammation (Calcagni and Elenkov, 2006). Administration of ASMq was found to increase the serum level of IL-2 and INF- β , thus, enhancing the secretion of the protective factors. On the contrary, the serum level of IL-6 was decreased by administration of ASMq. It is well known that inappropriate regulation of IL-6 may play a protective as well as a deleterious role in antigen-specific immune-mediated diseases, characterized by low-grade inflammation state (Jones et al., 2001; Kimura et al., 2003). IL-6 can also exert multiple stimulatory effects on cell growth and inflammation (Scheller et al., 2006), being involved in the initiation and the maintenance of the acute phase inflammatory response in the immunoregulation and non-immune events in cell types and tis-

issues outside the immune system (Kristiansen and Mandrup, 2005; Esper et al., 2006). ASMq also has caused an increase of macrophage phagocytic activity. Recently, Cheng et al. (2008) have studied the immunomodulatory effects of the purified glycyrrhiza polysaccharides (GP) on the macrophages. They have found that GP was able to increase the pinocytic activity, the production of nitric oxide (NO), interleukin-1 (IL-1), IL-6 and IL-12 in a dose dependent manner. They have also found that superoxide anion (O₂⁻) production by macrophage in the GP-treated mice was higher than that of cells from untreated mice. These results have allowed them to suggest that the beneficial therapeutic effects of GP may be attributed partly to its ability to modulate macrophage immune functions.

Administration of ASMq also caused an alteration of the serum level of immunoglobulins. It is interesting to observe that only the levels of IgG were increased in the combined stress conditions. Administration of ASMq was able to bring down the levels of IgG in the combined stress mice. As the concentration IgG is associated with various pathological changes such as inflammation, activation of complement, immune complex adhesion and immune deregulation, the decrease of IgG serum levels by AMSq can help attenuate these prejudicial effects.

5. Conclusions

The results of this study have demonstrated that ASMq, a herbal preparation used in the Traditional Uighur Medicine in the Xinjiang Province of China for the prevention of Abnormal Savda Syndrome, was able to modulate both cellular and humoral immunities in the combined stress mice model. The results of this study can substantiate, in part, the therapeutic value of this herbal prescription. However, more insight on the mechanism by which this preparation exerts on the immune system is necessary to support the claim of its therapeutic efficacy.

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